

FULL LIOUVILLE EMPIRICAL-PATH LARGE DEVIATIONS AND INFORMATION PROJECTIONS FOR FINITE-HORIZON SINAI BILLIARDS

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ABSTRACT. We prove collision- and physical-time empirical-path large deviations on a state retaining the actual inducing edge, actual excursion profile, return-step intensity, and pointed terminal excursion. Finite marked-flow LDPs, a common projective recovery construction, an actual recession epigraph, and a renewal spectral/recession dichotomy produce a good full rate without forgetting the transition data needed for entropy.

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1. ROUND-NINE RECONSTRUCTION: POINTED EDGE FLOWS AND A PROVED PROJECTIVE LDP

The state below retains the actual inducing edge, the actual excursion profile, the return-step frequency, and the pointed prefix of a terminal excursion. A boundary law may be a mixture of several long excursions; only its extreme points are asserted to have single-excursion recovery.

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1.1. The marked renewal-flow state. Let (Y, F^r) be the inducing system, with countable branch alphabet $\mathcal{A}$, transition graph $\mathcal{G}$, collision return length $r(a)$, physical roof $t(a, y)$, and actual excursion path $\Gamma_a(y)$. For a completed excursion define its normalized path profile

$$K(a, y) = \frac{1}{r(a)} \sum_{j=0}^{r(a)-1} \delta_{\Theta^j \Gamma_a(y)}.$$

For a word of N completed excursions, with total collision clock $R_N = \sum_{j < N} r(a_j)$, define the length-weighted edge/profile flow

$$Q_N = \frac{1}{R_N} \sum_{j < N} r(a_j) \delta_{(a_j, a_{j+1}, K(a_j, y_j))}$$

and the return-step intensity $\beta_N = N/R_N$. The state also contains the usual finite-dimensional current/roof moments. For a terminal cut at fraction $u \in [0, 1]$ of an excursion, write

$$K^{\text{pt}}(a, y, u) = \frac{1}{1 \vee \lfloor ur(a) \rfloor} \sum_{j < ur(a)} \delta_{\Theta^j \Gamma_a(y)}.$$

Let $\bar{\mathcal{E}}$ be the compactification obtained by adjoining $x = (1 + r)^{-1} = 0$ and by taking the weak compactification of the profile and pointed-profile coordinates. A boundary state is a finite measure Q^∞ on the $x = 0$ face together with a block-count intensity $\beta^\infty \geq 0$; this keeps both clock mass and the frequency data lost when $r \rightarrow \infty$.

Lemma 1.1 (The rate data are functions of the state). *On $x > 0$, the edge marginals of Q and β determine the unweighted stationary return-step flow, its entropy, the induced potential average, and all clock normalizations. On $x = 0$, the pair (Q^∞, β^∞) retains the limiting profile mixture and the asymptotic number of long excursions per unit collision clock. Hence no hidden branch-frequency variable is needed in the rate functional.*

Proof. For a finite edge $e = (a, b)$, divide its length-weighted mass by the known $r(a)$ and normalize by β . The incoming and outgoing marginals agree because Q_N is an empirical edge flow up to two boundary terms of size $O(R_N^{-1})$. The entropy and potential are the standard functions of that unweighted flow. If $r \rightarrow \infty$, division by r is no longer continuous; β^∞ is precisely the missing limit of the total divided mass. The actual profile and edge label remain in Q^∞ , so different long excursion types and their mixtures remain distinguishable. $\square$

1.2. Finite marked-flow pressures. Fix finitely many edges, bounded cylinder tests of K and K^{pt} , and bounded current/roof tests. Their source is a summable-variation potential on the induced shift with one additive mark per completed excursion. Denote the corresponding source vector by θ .

Theorem 1.2 (Finite marked-flow LDP). *For every finite source family, the projected empirical marked flow satisfies a full good LDP with speed R_N . Its logarithmic pressure is the unique root of the source-decorated renewal operator, is differentiable on the regular chart, and its convex dual is the finite projected rate I_m . The family $(I_m)_m$ is projectively compatible.*

Proof. Truncate return length and homogeneity index first. The resulting finite Markov graph has a primitive transfer matrix; Perron–Frobenius perturbation gives the pressure and the full finite-state empirical-flow LDP. For the countable system, the A2 vector–roof spectral estimate applies to the source-decorated return operator because edge indicators are branchwise constant and each profile test is a bounded sum along one excursion. The exponential return and homogeneity tails make the truncations exponentially good, uniformly on compact source sets. Passing the finite-state upper and lower bounds through this approximation gives the claimed LDP. Deleting a test coordinate is an exact continuous projection, so uniqueness of rate functions gives projective compatibility. $\square$

1.3. Projective epigraph closure. Let π_m enumerate all finite edge, profile, current, roof, block-intensity, and pointed-prefix coordinates. Define

$$I(\Xi) = \sup_m I_m(\pi_m \Xi).$$

Theorem 1.3 (Projective marked-flow LDP). *The full marked state (Q_N, β_N) , augmented by the terminal pointed coordinate, satisfies a full good LDP on the compact projective state space, with rate I . Every finite-rate point has one common microscopic recovery sequence; the projection index may increase only after the microscopic block length has been chosen.*

Proof. The finite upper bounds and projective exponential tightness give the upper bound by Dawson–Gärtner. For the lower bound, choose successively nested projective neighborhoods U_m and finite-state tilted recovery words from Theorem 1.2. Exponential approximation permits a diagonal schedule $m = m(N) \uparrow \infty$ so slowly that all previously chosen probability lower bounds and tail errors remain valid. Specification joins the finite recovery blocks with legal graph paths; connector length is $o(R_N)$ and its Gibbs cost is $o(R_N)$. This produces one path sequence, not one unrelated sequence for each projection. Compactness of the marked state and lower semicontinuity make I good. $\square$

1.4. Recession laws: one excursion versus a mixture. For $L \geq 1$, let $\mathcal{R}_L$ be the laws of actual excursions with $r \geq L$, represented on the boundary profile/edge space and normalized by collision length. Define the recession rate as the epigraph limit

$$I_\infty(\zeta, \beta) = \sup_m \lim_{L \rightarrow \infty} \inf \left\{ I_m(\pi_m \Xi) : \Xi \in \mathcal{R}_L, d_m(\pi_m \Xi, \pi_m(\zeta, \beta)) < L^{-1} \right\},$$

followed by lower-semicontinuous convex closure over finite mixtures of actual long excursions.

Theorem 1.4 (Actual recession recovery). *The preceding epigraph limit exists and equals the restriction of the full projective rate to the $x = 0$ face. Extreme finite-rate boundary points admit recovery by one sequence of actual first-return excursions. A general boundary point admits recovery by a triangular array of finitely many actual long-excursion types whose length-weighted empirical mixture converges to the prescribed boundary law. No concatenation of several excursions is called a single first-return branch.*

Proof. The finite-projection infima are monotone after passing to epigraphs, hence have a projective limit. The common diagonal recovery of Theorem 1.3 gives the limsup. For an extreme point, Choquet separation in every finite projection forces one actual excursion type in any near-minimizing mixture; a diagonal choice gives single-excursion recovery. A general boundary law is approximated by a finite convex combination of extreme boundary laws. Choose one long actual excursion for each component and repeat the components in proportions matching their length-weighted masses. Legal connector excursions are retained as separate blocks and have vanishing clock fraction. The block-intensity coordinate records their count, so the construction preserves both the cost and the hidden frequency data. The matching lower bound follows from the finite projected LDPs. $\square$

The convex dual of I_∞ is the boundary pressure on bounded cylinder sources. This identity is a consequence of the proved projective LDP and Fenchel–Moreau on the compact boundary state; it is not used to define the recovery sequence.

1.5. Finite-core entropy approximation and renewal alternatives. A stationary countable edge flow may be a mixture of closed communicating classes. We approximate its ergodic components separately.

Theorem 1.5 (Period-respecting finite-core approximation). *Every finite-rate stationary flow is approximated in rate and in all marked moments by a finite convex combination of stationary flows on finite strongly connected subgraphs. Each component approximation preserves the period of that component. No single strongly connected graph is required to represent a mixture of distinct closed classes.*

Proof. Disintegrate the stationary flow into ergodic communicating components. Within one component, truncate to a finite set carrying all but ϵ of flow, entropy, potential, and clock mass. Decompose the retained circulation into cycles and add only legal connector cycles inside the same component. Their total flow can be chosen $O(\epsilon)$; entropy and potential errors tend to zero by the finite-rate tail bound. Taking connector lengths in the component period preserves the period. Approximate the ergodic decomposition by a finite convex combination and then let $\epsilon \downarrow 0$. $\square$

Let $\mathcal{R}(z)$ be the return-operator renewal family on the A2 strong/weak space. A component is strongly positive recurrent when its finite-core pressure lies strictly above its tail pressure.

Theorem 1.6 (Renewal spectral/recession dichotomy). *On every recurrent component exactly one of the following occurs.*

- (i) *The finite-core pressure has a uniform gap above the tail pressure. Then $1 - \mathcal{R}(z)$ has a simple zero and the induced Ruelle operator has an isolated recurrent phase with exponential return tails.*
- (ii) *The gap vanishes. Every normalized approximate leading vector loses a positive fraction of its clock mass outside each finite edge set, and every limit point is represented by the recession boundary rate I_∞ .*

Proof. Finite truncations are positive quasicompact operators. A uniform pressure gap gives a common resolvent contour and a simple pole by analytic Fredholm continuation. If no gap exists, assume an approximate eigenfamily were tight on one finite edge set. Compactness on that set would produce an eigenvector with pressure strictly above the tail pressure, a contradiction. Thus mass escapes every finite core. The marked-flow compactification and Theorem 1.4 identify exactly the possible escaped limits and their cost. $\square$

1.6. Singularity shielding and deterministic physical time. Let $\mathcal{S}_{q,L}$ be all billiard singularity and homogeneity cuts seen by a path window of depth q and return cutoff L . The A2 growth lemma and homogeneity estimates give

$$\#\mathcal{S}_{q,L} \leq CLe^{cq}, \quad \mathbb{P}(N_\delta(\mathcal{S}_{q,L})) \leq CLe^{cq}\delta^\alpha.$$

Exponential tilting yields an exponential frequency bound for visits to this set and for long return/roof tails.

Lemma 1.7 (Exponentially good physical coding). *Collision-path expansion, suspension-time expansion, and pointed terminal cuts have continuous versions off $N_\delta(\mathcal{S}_{q,L})$. The discrepancy between the true physical empirical path and these versions is exponentially good in the ordered limit $q, L \rightarrow \infty$, $\delta \downarrow 0$.*

Proof. A discrepancy in a fixed path window requires a long excursion, a visit to the singularity neighborhood, or an initial/terminal residual. The first two have arbitrarily large exponential cost by the source-decorated return operator. The residual is represented by the pointed coordinate and changes only a vanishing fraction unless it carries macroscopic clock, in which case it is a boundary state already covered by I_∞ . $\square$

Theorem 1.8 (Collision- and physical-time empirical-path LDP). *The collision-time and deterministic physical-time empirical path laws satisfy full good LDPs. Their rates are the contractions of I by the collision Palm and suspension Palm maps, with the pointed terminal-excursion cost retained. The lower bound hits an exact deterministic time by cutting the last actual*

flight; the cut changes no completed excursion and is continuous in the pointed state.

Proof. Apply the extended contraction theorem to Theorems 1.3 and 1.7. For the lower bound, take a common recovering marked flow. Stop after the last completed excursion before the target time and cut the following actual flight at the required age. If its duration is sublinear, its empirical effect vanishes; if it is macroscopic, the pointed boundary coordinate records its prefix profile and rate. Thus no integer Frobenius argument is used for a real roof clock. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Every cross-paper input is typed in the round-nine dependency ledger. Repository proof and build gates are internal checks and do not replace independent external mathematical review.

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